

BILL OF MATERIALS

Component Name	Quantity
16×2 LCD (LCD1)	1
Arduino Uno (MOD1)	1
DS18B20 temperature sensor (S1)	1
4×4 keypad (MOD2)	1
Servo motor MG996R (M1)	1
5V relay (RL1)	1
Buzzer (B1)	1
1N4007 diode (D1)	1
5mm LED (D2)	1
Push-to-off switch (SW2)	1
Slide switch SPDT (SW1)	1
NPN BC547 transistor (T1)	1
Resistor 1kΩ (R1)	1
Resistor 4.7kΩ (R2)	1
10kΩ potentiometer	1
Immersion rod	1
Gas cylinder knob panel	1
Heater 1000kΩ	1
AC socket	1

servo motor, and relay to ensure timely safety response and real-time monitoring.

The system operates in two modes selected using a slide switch:

1. Mode 1 (gas safety mode):

Uses a servo motor to close a gas cylinder knob when milk overheats

2. Mode 2 (heater safety mode):

Uses a relay to switch off a heater rod when water overheats

Mode 1: Gas cylinder knob control (slide switch toward low side). When the slide switch is set to the low position, the system enters gas safety mode.

In many households, milk is heated until it reaches a desired temperature. However, delay in switching off the gas often leads to overflow. This system prevents such incidents.

A threshold temperature is set using the keypad before heating begins. The DS18B20 sensor monitors the temperature continuously. When the measured value exceeds the threshold, the servo motor rotates from its initial position (about 20°) to 180° for a one-second duration, simulating the closing

of the gas knob.

The servo then returns to its initial position while the gas knob remains off. At the same time, an LED and a buzzer are activated to alert the user. The alert can be turned off using a push-button switch.

The LCD displays both current and set temperatures for easy monitoring.

Mode 2: Heater rod safety control (slide switch toward high side). When the slide switch is set to the high position, the system switches to heater safety mode.

This mode reduces risks associated with heating water using immersion rods. Manual checking of water temperature can result in burns or electric shock if the water becomes excessively hot.

In this setup, the sensor is placed in the water and a desired temperature is set using the keypad. As the immersion rod heats the water, the Arduino continuously

monitors the temperature. Once the temperature exceeds the preset limit, the 5V relay energises, cutting off the power supply to the heater and preventing overheating.

At the same time, an LED and a buzzer are activated to indicate that the set limit has reached. The system remains in this state until the temperature drops below the threshold, ensuring improved safety.

Software

The program is developed using the Arduino IDE. If not already installed, it can be downloaded from the official Arduino website.

The required libraries are installed using the Library Manager: LiquidCrystal.h, DallasTemperature.h, Keypad.h, and Servo.h.

The code begins with inclusion of the necessary libraries and configuration of the display for a 16-pin parallel LCD. Pins for switches, relay, and other components are defined and initialised.

```

1 //RAKESH JAIN PROGRAM FOR Innovative Temperature-Based Dual Safety System 1
2 #include <LiquidCrystal.h>
3 #include <OneWire.h>
4 #include <DallasTemperature.h>
5 #include <Keypad.h>
6 #include <Servo.h>
7 // Define Pins
8 #define ONE_WIRE_BUS A2 // DS18B20 data pin
9 #define ALERT_PIN A3 // LED + Buzzer
10 #define SERVO_PIN A4 // Servo motor
11 #define MODE_SWITCH A5 // Slide switch input
12 #define RELAY_PIN 1 // Relay (using TX pin)
13 // DS18B20 Setup
14 OneWire oneWire(ONE_WIRE_BUS);
15 DallasTemperature sensors(&oneWire);
16 // Servo Setup
17 Servo myServo;
18 int basePos = 20;
19 int activePos = 180;
20 // LCD Setup
21 LiquidCrystal lcd(2, 3, 4, 5, 6, 7);
22 // Keypad Setup
23 const byte ROWS = 4;
24 const byte COLS = 4;
25 char keys[ROWS][COLS] = {
26   {'1', '2', '3', 'A'},
27   {'4', '5', '6', 'B'},
28   {'7', '8', '9', 'C'},
29   {'*', '0', '#', 'D'}
30 };
31 byte rowPins[ROWS] = {8, 9, 10, 11};
32 byte colPins[COLS] = {12, 13, A0, A1};
33 Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);
34 // Variables
35 int thresholdTemp = 95; //we can set any thersold temperature according 1

```

Fig. 2: Source code snippet